

AGILE SOFTWARE ENGINEERING (CS303)

10 Marks Question Bank

UNIT 1 – Agile Fundamentals

1. Explain the Agile Manifesto and discuss the four core values in detail.
2. Describe the twelve Agile Principles with suitable examples.
3. Compare Waterfall and Agile methodologies in terms of process, flexibility, risk, and customer involvement.
4. Discuss the advantages and challenges of Agile Software Development.

UNIT 2 – Scrum Framework

1. Explain the Scrum Framework and its lifecycle with a neat diagram.
2. Discuss the roles and responsibilities of Product Owner, Scrum Master, and Development Team.
3. Explain Sprint Planning and Sprint Review processes in Scrum.
4. Differentiate Product Backlog and Sprint Backlog with examples.

UNIT 3 – Agile Requirements

1. Explain User Stories and describe how they are written using the INVEST principle.
2. Discuss Story Points and estimation techniques used in Agile projects.
3. Explain Agile Requirement Engineering and its importance.
4. Describe Acceptance Criteria with suitable examples.

UNIT 4 – Extreme Programming (XP)

1. Explain Extreme Programming (XP) and its core practices.
2. Discuss Pair Programming and its advantages and disadvantages.
3. Explain Test Driven Development (TDD) with the Red-Green-Refactor cycle.
4. Describe Refactoring and explain its importance in software quality improvement.

UNIT 5 – Agile Metrics and CI/CD

1. Explain Agile Metrics and discuss their role in project monitoring.
2. Describe Burn-down Charts and explain how they are used in Scrum projects.
3. Explain Continuous Integration and Continuous Delivery with examples.
4. Discuss the architecture and workflow of a CI/CD Pipeline.

Exam Preparation Tips

Focus on Agile Manifesto, Scrum Framework, User Stories, Story Points, Extreme Programming, TDD, Burn-down Charts, and CI/CD Pipelines as they are frequently asked 10-mark university examination questions.